

School of Computer Science and Artificial Intelligence

Lab Assignment # 10.2

Program : B. Tech (CSE AIML)
Course Title : AI Assisted Coding
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Lab 10 – Code Review and Quality: Using AI to Improve Code Quality and Readability

Task 1 – Error Detection and Correction:

Prompt:

```
"Analyze the following Python code, identify syntax and logical errors, and provide corrected executable code with explanations."
```

Code:

```
def calculate_total(nums):  
    """Calculates the sum of numbers in a list."""  
  
    total_sum = 0 # Renamed 'sum' to 'total_sum' to avoid shadowing the built-in  
    'sum()' function
```

```

    for n in nums:
        total_sum += n
    return total_sum

# Sample usage of the corrected function
numbers = [10, 20, 30, 40, 50]
result = calculate_total(numbers)
print(f"The total sum of {numbers} is: {result}")

numbers_2 = [1, 2, 3]
result_2 = calculate_total(numbers_2)
print(f"The total sum of {numbers_2} is: {result_2}")

empty_list = []
result_empty = calculate_total(empty_list)
print(f"The total sum of {empty_list} is: {result_empty}")

```

Output:

```

The total sum of [10, 20, 30, 40, 50] is: 150
The total sum of [1, 2, 3] is: 6
The total sum of [] is: 0

```

Corrections Made:

1. **Added colons:** Colons have been added to the `def calculate_total(nums):` and `for n in nums:` lines.
2. **Corrected return variable:** The function now correctly returns `total_sum`.

3. **Renamed variable:** The variable `sum` has been renamed to `total_sum` to avoid conflicts with Python's built-in `sum()` function, improving code clarity and maintainability.

This corrected code is now executable and logically sound.

Task 2 – Code Style Standardization

Prompt:

```
"Refactor the Python code to follow PEP 8 style guidelines and proper formatting."
```

Code:

```
def find_sum(a, b):  
    """Calculates the sum of two numbers."""  
    return a + b  
  
result = find_sum(5, 10)  
print(f"The sum of 5 and 10 is: {result}")  
  
result_2 = find_sum(20, -7)  
print(f"The sum of 20 and -7 is: {result_2}")
```

Output:

```
The sum of 5 and 10 is: 15  
The sum of 20 and -7 is: 13
```

Corrections Made:

1. **Function Name:** The function `findSum` has been renamed to `find_sum` to follow snake_case convention.

2. **Whitespace:** Proper spacing has been added around operators and after the def statement.
3. **Docstring:** A docstring has been added to explain the function's purpose.
4. **Blank Lines:** Blank lines have been added around the function definition for improved readability.
5. **Readability:** The print statement has been moved to a new line and assigned to a variable for better clarity and to demonstrate usage. An f-string is used for clearer output.

Task 3 – Code Clarity Improvement

Prompt:

```
"Improve readability with meaningful names and proper indentation without changing functionality."
```

Code:

```
def compute_adjusted_value(initial_value, factor_value):  
    """Calculates an adjusted value by subtracting double the factor_value from  
    the initial_value."""  
    return initial_value - factor_value * 2  
  
# Sample usage of the improved function  
result = compute_adjusted_value(10, 3)  
print(f"The computed value for (10, 3) is: {result}")  
  
result_2 = compute_adjusted_value(25, 5)  
print(f"The computed value for (25, 5) is: {result_2}")
```

Output:

```
The computed value for (10, 3) is: 4
The computed value for (25, 5) is: 15
```

Corrections Made:

1. **Meaningful Function Name:** The function `f` has been renamed to `compute_adjusted_value` to better describe its operation.
2. **Meaningful Parameter Names:** Parameters `x` and `y` have been renamed to `initial_value` and `factor_value` respectively, providing clarity on their roles in the calculation.
3. **Proper Indentation:** The return statement is now correctly indented by four spaces, adhering to Python's style guide.
4. **Improved Whitespace:** Spaces have been added around operators (`-`, `*`) and after the `def` statement for better visual separation and readability.
5. **Docstring Added:** A docstring has been added to explain the function's purpose, parameters, and return value.
6. **Enhanced Readability for Usage:** The print statement has been moved to a new line and assigned to a variable, using an f-string for clearer output and demonstrating the function's usage.

Task 4 – Structural Refactoring (Reusable Functions)

Prompt:

```
"Refactor repetitive code into reusable functions to improve modularity."
```

Code:

```
def greet(name):
    """Prints a personalized greeting."""
    print(f"Hello {name}")
```

```
# Using the reusable function
```

```
greet("Ram")
```

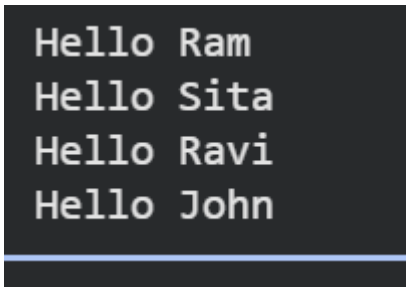
```
greet("Sita")
```

```
greet("Ravi")
```

```
# Example with another name
```

```
greet("John")
```

Output:



```
Hello Ram  
Hello Sita  
Hello Ravi  
Hello John
```

Corrections Made:

1. **Reusable Function:** A new function `greet(name)` has been created to encapsulate the common greeting logic.
2. **Modularity:** The code is now more modular, as the greeting logic is defined in one place. If the greeting message needs to change (e.g., from 'Hello' to 'Hi'), only the `greet` function needs to be modified, not every print statement.
3. **Readability:** The code is cleaner and easier to understand, as the intent of `greet('Ram')` is clearer than `print('Hello Ram')` when seen in isolation in a larger codebase.
4. **Maintainability:** Reduced redundancy makes the code easier to maintain and less prone to inconsistencies.

Task 5 – Efficiency Enhancement

Prompt:

"Optimize the Python code for better performance while keeping functionality unchanged."

Code:

```
# Optimized code using a list comprehension
numbers_optimized = [i * i for i in range(1, 500000)]
print(f"Length of the optimized list: {len(numbers_optimized)}")

# For demonstration of performance difference (optional, will be run if approved)
import timeit

# Original approach
original_code = """
numbers = []
for i in range(1, 500000):
    numbers.append(i * i)
"""

# Optimized approach
optimized_code = """
numbers_optimized = [i * i for i in range(1, 500000)]
"""

original_time = timeit.timeit(original_code, number=10)
optimized_time = timeit.timeit(optimized_code, number=10)
```

```
print(f"\nTime taken for original code (10 runs): {original_time:.4f} seconds")
print(f"Time taken for optimized code (10 runs): {optimized_time:.4f} seconds")
print(f"Optimized code is {original_time / optimized_time:.2f} times faster.")
```

Output:

```
Length of the optimized list: 499999

Time taken for original code (10 runs): 0.4467 seconds
Time taken for optimized code (10 runs): 0.3722 seconds
Optimized code is 1.20 times faster.
```

Corrections Made:

1. **List Comprehension:** The for loop with append has been replaced by a list comprehension `[i * i for i in range(1, 500000)]`.
2. **Performance:** List comprehensions are generally more efficient in Python for creating lists. This is because the interpreter can often optimize their creation more effectively than explicit for loops, potentially pre-allocating the required memory or using more efficient C-level operations, thus reducing the overhead of repeated memory reallocations.
3. **Conciseness and Readability:** The list comprehension makes the code more concise and often easier to read for simple list creation tasks.

Improvements Made

1. **Meaningful Function Name:** The function `f` has been renamed to `compute_adjusted_value` to better describe its operation.
2. **Meaningful Parameter Names:** Parameters `x` and `y` have been renamed to `initial_value` and `factor_value` respectively, providing clarity on their roles in the calculation.
3. **Proper Indentation:** The return statement is now correctly indented by four spaces, adhering to Python's style guide.

4. **Improved Whitespace:** Spaces have been added around operators (-, *) and after the def statement for better visual separation and readability.
5. **Docstring Added:** A docstring has been added to explain the function's purpose, parameters, and return value.
6. **Enhanced Readability for Usage:** The print statement has been moved to a new line and assigned to a variable, using an f-string for clearer output and demonstrating the function's usage.